

Combining Heterogeneous Spatial Data From Distributed Sources

M. Howard Williams and Omar Dreza

School of Math & Comp. Sc., Heriot-Watt Univ., Riccarton, Edinburgh,
EH14 4AS UK,

Abstract

The general problem of retrieval and integration of data from a set of distributed heterogeneous data sources in response to a query has a number of facets. These include the breakdown of a query into appropriate sub-queries that can be applied to different data sources as well as the integration of the partial results obtained to produce the overall result. The latter process is non-trivial and particularly dependent on the semantics of the data. This paper discusses an architecture developed to enable a user to query spatial data from a collection of distributed heterogeneous data sources. This has been implemented using GML to facilitate data integration. The system is currently being used to study the handling of positional uncertainty in spatial data in such a system.

Keywords: Spatial databases, XQL, GML, distributed databases, heterogeneous data.

1 Introduction

The problem of retrieving and integrating information from a set of heterogeneous data sources involves a number of complex operations (El-Khatib et al. 2002). These include the breakdown of a query into appropriate sub-queries that can be applied to different data sources and the integration of the partial results obtained to produce the overall result (MacKinnon et al. 1998). Initially the problem was more difficult to deal with but progress in communication and database technologies has facilitated solutions. The present situation is characterized by a growing number of

of applications that require access to data from a set of heterogeneous distributed databases (Elmagarmid and Pu 1990).

This need for access to distributed data is certainly true of spatial data where there has been growing interest in retrieval and integration of data from distributed sources. One development in the USA is the National Spatial Data Infrastructure (NSDI) which developed under the coordination of the Federal Geographic Data Committee (FGDC 2003). This defined the technologies, policies, and people necessary to promote sharing of geospatial data throughout all levels of government. It provides a base or structure of practices and relationships among data producers and users that facilitates data sharing and use. It is a set of actions and new ways of accessing, sharing and using geographic data that enables far more comprehensive analysis of data to help decision-makers.

The implementation of Internet GIS requires not only network infrastructures to distribute geospatial information, but also software architectures to provide interactive GIS functions and applications. This paper discusses an architecture developed to demonstrate the integration of heterogeneous spatial data from distributed data sources to support particular decisions.

2 Query process

In order to provide a user-friendly interface that is easily understandable to users, one may develop graphical or natural language interfaces to such a system. A query at this level may involve data from different data sources. Thus the first task is to map a query from this general form into sub-queries addressed to specific data sources and expressed in an appropriate query language. For example, a query such as “What roads may be affected in region (xmin, ymin, xmax, ymax) if the level of lake x rises by 3 metres?” may require three different sets of data - a map of the lake, the road network and the DTM data sources. Thus this general query must be translated into three specific sub-queries, each formulated in the language that is understandable by the related database manager.

In order to implement this, each data provider needs to develop a wrapper (Zaslavsky et al. 2000) whose purpose is to translate a query from the server language to that of the data source and transform the results provided by the data source to a common format understandable by the server.

As this part of the process is not the main focus of our work, a simple approach has been adopted for our implementation.

3 Data integration process

Because of the autonomy of the data providers, the data generated from the three data sources mentioned in the previous section will in general have different syntax (i.e. different schemas), and different semantics. In order to integrate them it is necessary to convert them to a common format. For this purpose GML (GML 2001) was chosen. GML has the following properties:

- It is an XML encoding of geography,
- It enables the GI community to leverage the world of XML technology,
- It supports vector mapping in standard Web applications,
- It provides complex features, feature collections and feature associations.

4 Architecture

The architecture described here has been developed and implemented to support access to a distributed collection of heterogeneous spatial data sources. The basic idea behind this architecture is that the user should be able to pose a query which requires data from different sources and that the system will fetch the data and combine it to produce an answer without the user being aware of the different data sources involved. The architecture is based on a client/server approach, with three levels - the client level, the server level, and the data provider level. A system based on this architecture has been developed using Java.

4.1 The Client User Interface

Currently the system uses a very simple Client User Interface (CUI) with a set of templates for fixed scenarios. Since the main focus of our work is on the handling of positional uncertainty this provides a simple and easy to use interface although limited for general use.

The CUI is designed to allow the user to specify the keys of his/her query such as the region name, coordinates, etc. For a scenario such as flooding, the user can specify the affected entities (roads, land parcels, buildings, etc), the entity causing the flooding, etc. For a scenario such as utilities maintenance, the user can specify the utilities that may be affected

by the maintenance (water pipe, gas pipe, etc.), the maintaining utility, and key parameters (such as digging width, measurement unit, etc.).

Fig. 1 shows an example of the CUI for a flooding scenario query. The CUI presents the template to the user and obtains from the user the values of the parameters. These are then used to generate a query that is sent to the Server.

The client also has a browser component for viewing the results retrieved by the Server. As shown in fig. 5 this browser contains GIS tools to help the user to zoom, pan, and select specific features. It also displays the scale of the viewed results and can send the results to a printer to produce hardcopy.

Cond.	Value	Color	Legend
=	100	Red	Definitely flooded
>	67	Orange	High probability
>	33	Yellow	Medium probability
>	0	Light Green	Low probability
=	0	Green	Definitely not flooded

Fig. 1. Client User Interface

4.2 The Server

The Server works as an information mediation service where user queries against multiple heterogeneous data sources are communicated via middleware (*mediator*) which is responsible for query rewriting, dispatching query fragments to individual sources, and assembling individual query results into a composite query response (Levy et al. 1996; Papakonstantinou et al. 1996; Tomasic et al. 1998; Wiederhold 1992).

The proposed architecture of the server contains a number of processes used to analyse the query sent by the client as illustrated in Fig. 2. The server has three main purposes. The first is to break down the query passed

from the client into a simple set of sub-queries, which can be understood by the data provider taking account of the purpose of the client query. This is performed by the KB component that generates the separate sub-queries required to satisfy the general query.

The second purpose is to determine appropriate data sets, which can be retrieved from the data providers and which will satisfy the user query parameters (scale, locations).

Finally, the server must integrate the spatial results sent back to the server from the data providers. Once again the KB component is used to perform this integration. This task will be further illustrated in section 5.

The server metadata consists of summary extracts derived from the provider metadata. It contains information about the data sets from all data providers known to the server. This information includes data provider URL address, the scale of the data set, the spatial location covered by the data set, etc., and is used to determine whether a data set is relevant to a specific sub-query and, if it is, how to perform a connection to the data provider. The following example shows part of the server metadata that is retrieved for the lake flood scenario.

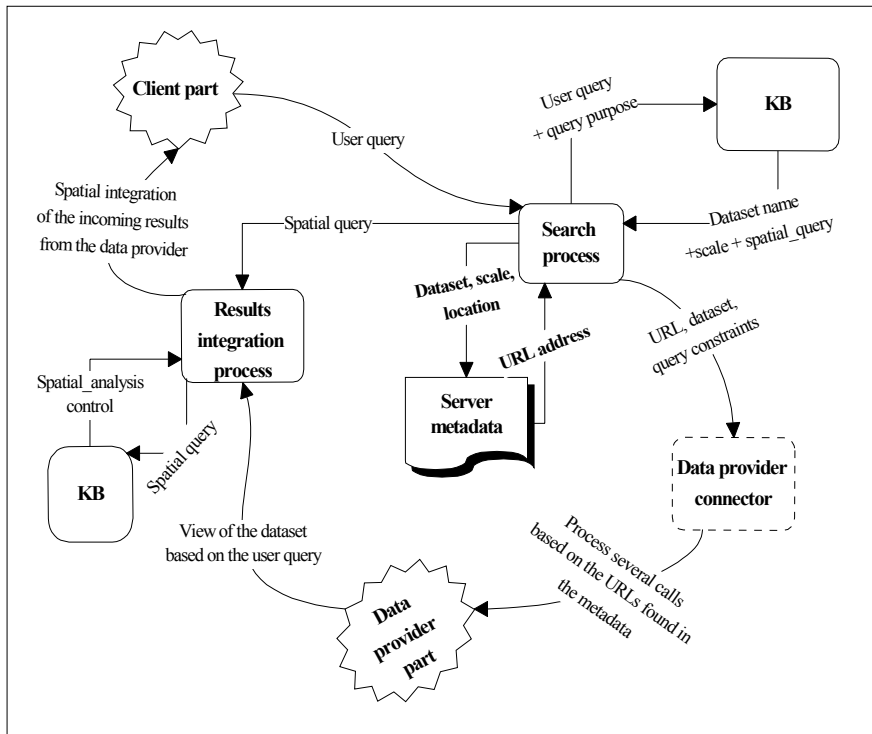


Fig. 2. Server Architecture

```
<DATASET_NAME>LAKE
  <SCALE>25000</SCALE>
  <URLADDRESS>
    lab152pc2.cee.hw.ac.uk:1115
  </URLADDRESS>
  <REGION>EDINBURGH</REGION>
</DATASET_NAME>
<DATASET_NAME>ROAD
  <SCALE>25000</SCALE>
  <URLADDRESS>
    PCDB3.macs.hw.ac.uk:1115
  </URLADDRESS>
  <REGION>EDINBURGH</REGION>
</DATASET_NAME>
<DATASET_NAME>DTM
  <SCALE>25000</SCALE>
  <URLADDRESS>
    lab152pc3.cee.hw.ac.uk:1115
  </URLADDRESS>
  <REGION>EDINBURGH</REGION>
  <BOUNDARY>
    <WEST>328500.0</WEST>
    <EAST>331530.0</EAST>
    <NORTH>652260.0</NORTH>
    <SOUTH>655480.0</SOUTH>
  </BOUNDARY>
</DATASET_NAME>
```

The KB_XML file contains a set of knowledge relating to the specific data sets that can be used. Each set contains a spatial query related to a particular problem and the optimum scale that can be used in this particular scenario.

The KB component takes the parameters of the query provided by the client (e.g. affected entities, etc.) and converts them to an XML query, which is used to fetch the KB_XML file. This query is developed using the XQL standard. More information on this and the XCool library can be found in (Robie 1999).

The following is an example of the results retrieved from the KB_XML file with a flooding scenario XML query.

```
<DATASET scenario = "flood">
  <NAME>LAKE</NAME>
  <Optimum_SCALE>
```

```
25000
</Optimum_SCALE>
<Spatial_Query>
  SELECT * FROM LAKE WHERE LAKE.name = X
</Spatial_Query>
<LINK>DTM</LINK>
</DATASET>
<DATASET scenario = "flood">
  <NAME>ROAD</NAME>
  <Optimum_SCALE>
    25000
  </Optimum_SCALE>
  <Spatial_Query>
    SELECT * FROM ROAD WHERE ROAD.name = X
  </Spatial_Query>
</DATASET>
<DATASET scenario = "flood">
  <NAME>DTM</NAME>
  <Optimum_SCALE>
    25000
  </Optimum_SCALE>
  <Spatial_Query>
    SELECT * FROM REGION WHERE REGION.x >= Xmin And
    REGION.x <= Xmax And REGION.y >= Ymin And REGION.y <=
    Ymax
  </Spatial_Query>
</DATASET>
```

The Result integration process is responsible for integrating the data views sent to it from the various data providers corresponding to sub-queries based on the user query. The result integration process is responsible for integrating the spatial data views to satisfy the client query. This integration is based on a KB, which contains procedures for integrating data for specific scenarios. For example, for the scenario used to calculate the effect of a flood on a lake, the lake has to be overlapped with the DTM data. Knowing the height of the lake we can generate the contour for the flood boundaries. Then from overlaying the generated contour and the road, the affected roads can be determined.

4.3 Data provider

The main tasks of the data provider part of the architecture are to answer the server query and produce results to the server spatial query. When the server requires a data provider, it connects to the data provider process us-

ing RMI. Each data provider as shown in Fig. 3, contains two major components: the wrapper, and the Metadata.

The wrapper contains three processes. The first controls the retrieval of the data set. This is used to check for the location of data set by querying the data provider metadata. The following is an example of the part of the XML file, which relates to the DTM data provider metadata:

```
<?xml version="1.0"?>
  <DPMETADATA>
    <DATASET_NAME>DTM
    <SCALE>1:50,000</SCALE>
    <FILE_NAME>ID_points2</FILE_NAME>
  <DIRECTORY>
    H:\www\DTMdataProvider\DTM\
  </DIRECTORY>
  <REGION>Edinburgh</REGION>
  <HUNCERTAINTY> 3 </HUNCERTAINTY>
  <VUNCERTAINTY> 3 </VUNCERTAINTY>
</DATASET_NAME>
</DPMETADATA>
```

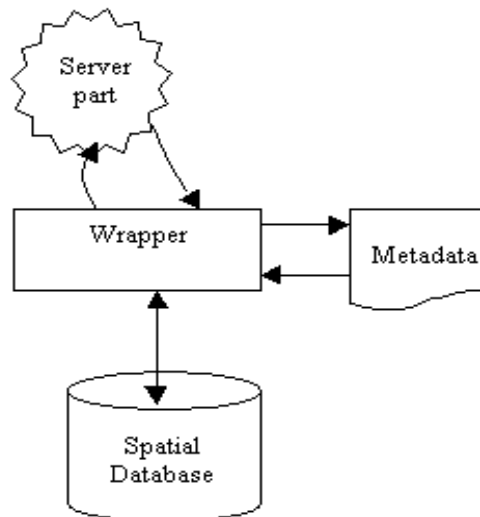


Fig. 3. Data provider

The second process is responsible for applying the spatial query to the spatial database. For example, if the spatial database used is MapInfo, the retrieval process is written in the MapBasic language provided by MapInfo. This could easily be extended to handle other spatial databases. The third process retrieves the results generated by the query. It converts the TAB file generated from the spatial query to a GML file or a text file (in the case of DTM data). This process is implemented using the MITAB library (MITAB 2002). Fig. 4 shows part of the GML file representing the retrieved road network data.

Finally, the uncertainty associated with the exact locations of features is

```
<?xml version="1.0"?>
<FeatureCollection>
  <SpatialReferenceSystem srsName="">
    <Projected name="">
      <LinearUnit>
        <Name>unspecified</Name>
      </LinearUnit>
      <Geographic name="">
        <Datum>
          <DatumName>unspecified</DatumName>
          <Spheroid>
            <SpheroidName>unspecified</SpheroidName>
          </Spheroid>
        </Datum>
      </Geographic>
    </Projected>
  </SpatialReferenceSystem>
  <Uncertainty> 3.0 </Uncertainty>
  <Feature fid="1" featureType="line" name="">
    <Description>1 Edinburgh Road1 </Description>
    <Property name="Edinburgh" Type="String"/>
    <Polygon name="Road1" srsName="">
      <LineString name="line" srsName="">
        <Clist>
          329167.8580,655677.3579 329176.0383,655571.1892
          329162.0340,655366.3166 329185.8037,655184.3520
          329193.8983,655093.3545 329142.1628,654857.9885
          329128.0728,654668.2872 328925.3267,654326.1183
          328882.7140,653817.6995 328815.6889,653605.0299
          328755.8159,653468.2469 328499.6635,653186.5517
          327820.3729,652660.4251
        </Clist>
      </LineString>
    </Polygon>
  </Feature>
```

Fig. 4. Part of GML file representing road network data.

determined and added to the results.

5 Using the approach

The approach described here has been implemented in a system developed to support access to a distributed collection of heterogeneous spatial data sources. To illustrate the operation of the system developed, consider the flooding scenario with a set of data sources. The results of a specific query of this type are shown in Fig. 5.

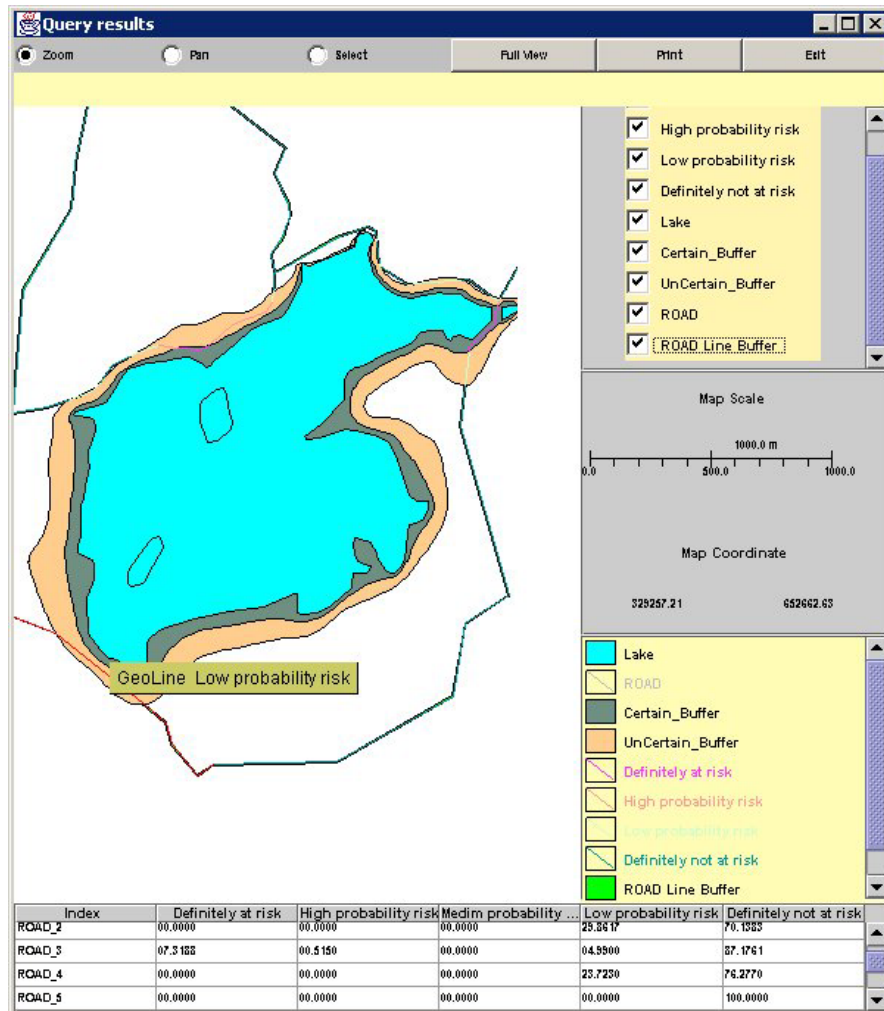


Fig. 5. Final results

6 Conclusion

This paper is concerned with the problem of combining spatial data from heterogeneous sources. An architecture that has been developed to handle queries involving multiple spatial data sources has been extended to incorporate different forms of uncertainty in the data.

A prototype has been implemented to realise this architecture. This was initially based on the use of XML as encoding standard for the metadata and the KB and to represent the data transferred in response to requests between different levels of the architecture. The system uses XQL as the query language to retrieve information that encoded in XML, such as metadata and KB.

GML has proved to be an excellent notation for representing vector-GIS data. It is simple and readable and based on XML schema. However, there are some limitations in handling large DTM files. Using GML schema, the data and its geometry can be stored in a single file, which makes it easier to represent data transferred between different spatial data sources.

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Reference

- El-Khatib HT, Williams MH, MacKinnon LM, Marwick DH (2002). Using a distributed approach to retrieve and integrate information from heterogeneous distributed databases. *Computer Journal*, 45(4):381-394.
- Elmagarmid AK, Pu C (1990) Introduction: special issue on heterogeneous databases (guest editors), *ACM Computing Surveys*, 22(3):175-178.
- FGDC (2003) FGDC, USGS, 590 National Center, Reston, VA 20192. Updated: Thursday, 27-Mar-2003. <http://www.fgdc.gov/nsdi/nsdi.html>
- GML 2001. Geography Markup Language (GML 2.0), OpenGIS® Implementation Specification, 20, OGC Document Number: 01-029. <http://opengis.net/gml/01-029/GML2.html>
- Levy A, Rajaraman A, Ordille J (1996) Querying Heterogeneous Information Sources Using Sources Descriptions. *Proceedings of the 22nd International Conference on VLDB*, pp. 251-262.

- Mackinnon LM, Marwick DH, Williams MH (1998) A model for query decomposition and answer construction in heterogeneous distributed database systems. *Journal of Intelligent Information Systems*, 11: 69-87.
- MITAB. MapInfo .TAB and .MIF/.MID Read/Write Library, <http://pages.infinet.net/danmo/e00/index-mitab.html>.
- Papakonstantinou Y, Abiteboul S, Garcia-Molina H (1996) Object Fusion in Mediator Systems. *Proceedings of the 22nd International Conference on VLDB*, pp. 413-424.
- Robie J (1999). XCOOL WWW document, <http://xcool.sourceforge.net>.
- Tomasic A, Raschid L, Valduriez P (1998) Scaling Access to Heterogeneous Data Sources with DISCO. *IEEE Transactions on Knowledge and Data Engineering*, 10(5):808-823.
- Wiederhold G (1992) Mediators in the Architecture of Future Information Systems. *IEEE Computer*, 25(3):38-49.
- Zaslavsky I, Marciano R, Gupta A, Baru C (2000) XML-based Spatial Data Mediation Infrastructure for Global Interoperability, *4th Global Spatial Data Infrastructure Conference* Cape Town, South Africa. <http://www.npaci.edu/DICE/Pubs/>.